

Extension without Amplification: How Fridays for Future Germany Reframed Climate Change on Instagram across Five Crises, 2018–2025

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1 Introduction

Climate change is widely regarded as one of the most pressing crises facing humanity. In August 2018, Greta Thunberg began a solitary school strike in front of the Swedish parliament under the slogan *Skolstrejk för klimatet* (school-strike for climate). What started as an individual protest quickly grew into Fridays for Future (FFF), one of the largest social movements in Germany, Europe and worldwide. While the core movement was almost exclusively composed of young people, especially pupils, it inspired allied initiatives such as Parents, Teachers and Scientists for Future. At its peak, the global climate strike of 20 September 2019 mobilized around four million people worldwide, including an estimated 1.4 million in Germany alone (Frankfurter Allgemeine Zeitung 2019).

From its emergence in late 2018 through roughly 2020, FFF was a prominent presence in public debate. Since then, however, a series of external crises, such as the COVID-19 pandemic, the war in Ukraine, and the resulting energy and economic pressures, has competed for public attention and issue salience. This raises the question of how a climate movement adapts its framing when other crises enter the public agenda.

This paper therefore asks whether FFF Germany adapted its communication frames in response to emerging crises and, if so, whether this adaptation occurred through the amplification of existing frames or their extension to new issues and inequality arenas. We address this question through a single-case longitudinal framing analysis of FFF Germany’s Instagram communication, the movement’s main channel for reaching its predominantly young audience. We divide the observation period into five crisis windows defined by exogenous shocks to track changes in the composition of the movement’s frames. An LLM-based scoring instrument quantifies the presence and intensity of fifteen theory-derived framing concepts in every post caption. Post-level regression then compares each crisis window with the movement’s founding phase, while accounting for confounding variables.

We find that FFF Germany extended its framing to the inequality arenas that became salient in the competing crises. Furthermore, the climate-justice master frame gained on climate-change framing in exactly the crises with a distributional or solidarity dimension. This extension, however, was not accompanied by an amplification of motivational framing. Mobilization instead follows the strike and election calendar, and does so as strongly in 2025 as in 2019.

The remainder of the paper is structured as follows. Section 2 outlines the theoretical framework, Section 3 describes the data and the crisis windows, Section 4 the scoring instrument and the analysis strategy, Section 5 presents the results, and Section 6 discusses their implications and limitations.

2 Literature and Theory

The understanding of collective action frames in this study builds on the framing perspective developed by Benford and Snow (2000), who conceptualize collective action frames as “action-oriented sets of beliefs and meanings that inspire and legitimate the activities and campaigns of a social movement organization” (Benford and Snow 2000, p. 614). They distinguish three core framing tasks. *Diagnostic* framing identifies a problem and attributes causality or responsibility. *Prognostic* framing proposes solutions or courses of action. *Motivational* framing provides reasons for action and encourages participation in collective action.

Mendelsohn et al. (2024) build on this framework and derive a more fine-grained annotation typology that distinguishes subcategories of the core framing tasks of Benford and Snow (2000). The three core tasks and the six subcategories used in this analysis are listed and defined in Table 1.

Table 1: Core framing tasks (Benford and Snow 2000) and their subcategories (Mendelsohn et al. 2024) with examples from the climate context.

Framing task	Subcategory
Diagnostic Identifies a problem and attributes blame or causality (<i>Who is responsible?</i>)	Problem Identification: Names and describes a specific problem or issue. Ex.: climate change
	Blame Attribution: Assigns blame for a societal problem to specific actors, institutions or systems. Ex.: the government, corporations, economic systems.
Prognostic Proposes a solution or a plan of action (<i>What should be done?</i>)	Solutions: Proposes specific solutions or measures to overcome a problem. Ex.: carbon taxes, reforestation, end of fossil fuels.
	Tactics: Describes strategies or tactics for achieving a movement’s goals. Ex.: protests, boycotts, petitions, contacting politicians.
Motivational Provides a call to arms and constructs vocabularies of motive to justify and sustain participation	Call to action: Tries to convince readers to join, participate in, or support a social movement. Ex.: “seid dabei”, “komm vorbei”
	Rationale: Delivers a reason why action is needed, such as urgency, moral obligation, or the consequences of inaction. Ex.: “es bleibt keine Zeit mehr”, “tomorrow is too late”

Benford and Snow (2000) note that frames are not static, but change in response to external events, internal movement dynamics, and counter-framing by opponents. For the present study, this dynamic character is particularly relevant because FFF operates in an issue environment increasingly shaped by crises beyond climate change. We therefore focus on two forms of frame adaptation identified by Benford and Snow (2000): *frame amplification*,

which strengthens the salience of existing beliefs and values, and *frame bridging* or *extension*, through which movements connect existing frames to ideologically congruent but previously unconnected issues or extend a frame beyond its primary domain.

Beyond the functional tasks of framing, this study examines the evolution of the broader master frame through which FFF interprets climate change. Master frames are particularly broad and flexible interpretive frameworks that can encompass and shape more movement-specific frames (Benford and Snow 2000). Within the climate movement context, della Porta and Parks (2014) identified a shift from a dominant *climate change* frame towards a *climate justice* frame. Prior to the 15th Conference of the Parties (COP15) in Copenhagen in 2009, climate change was predominantly framed as a scientific, ecological, or technocratic problem. Subsequently, the *climate justice* frame gained prominence by connecting climate change to questions of social and global justice. Its broader scope and the incorporated justice dimension make this master frame particularly suitable for linking climate change to other societal conflicts and inequality concerns.

To further refine the justice dimension and examine frame extension to other societal conflicts, we draw on the four inequality arenas identified by Mau et al. (2023) as central fault lines of contemporary conflict in Germany. The first, *today vs. tomorrow*, concerns primarily intergenerational inequalities, especially those associated with climate change. The second, *inside vs. outside*, covers membership and access to social and political resources and is particularly relevant in debates over migration. The third arena, *us vs. them*, deals with identity, race, gender, participation, representation, and the recognition of marginalized groups. Finally, *top vs. bottom* concerns economic inequalities, including differences in income and wealth and policies aimed at redistribution. The arenas provide a grid for asking not only whether FFF connects climate change to other inequalities, but which inequality a given crisis pulls the climate frame towards.

2.1 Research Hypotheses

The present study examines how FFF Germany’s framing on Instagram evolved between 2018 and 2025 and whether identifiable shifts in frame composition coincide with external crises.

H1 – Frame Scope: As competing crises increase competition for public attention and issue salience, Fridays for Future extends its climate frames to other inequality arenas.

Public attention is a scarce resource, and competing crises can challenge the salience of climate change as a political issue. We therefore expect FFF to extend its climate frames to the newly salient inequality arenas.

H2 – Frame Content: As crises with salient justice dimensions become more prominent, Fridays for Future increasingly uses the climate justice master frame relative to climate-change framing.

Many of the crises considered in this study contain a pronounced justice dimension. Economic crises raise questions about the unequal distribution of costs and resources, while war and energy insecurity raise questions of vulnerability, responsibility, and distributive consequences. We therefore expect the climate justice master frame to become more prominent relative to more general climate-change framing as such crises enter the public agenda.

H3 – Frame Function: As crisis conditions heighten the need for mobilization, Fridays for Future increases the relative presence of motivational frames within its discourse.

Crises can increase the perceived urgency of collective action. In such contexts, social movements may place greater emphasis on encouraging audiences to participate in collective action. We therefore expect motivational framing to become more prominent when crisis conditions increase the need to generate or sustain mobilization.

H4 – Frame Timing: Motivational framing increases in the period preceding collective action events, such as elections and global climate strikes.

Collective action events create an immediate need to encourage participation and coordinate action. We therefore expect motivational framing to become particularly prominent in the period immediately preceding elections and global climate strikes.

Exploratory findings later motivated a refinement of H1 and H4. These are specified in Section 4.2, after the analysis that motivated them.

3 Data

The data used in this study were obtained from Instagram via the Meta Content Library¹ with access provided under the European Union’s Digital Services Act (DSA). We retrieved all German-language Instagram posts published between 2018 and July 2026 whose caption contained the phrase “fridaysforfuture” or “fridays for future” (case-insensitive). Although the online search interface returns the full corpus of matching posts, only a subset of these can be downloaded. During the time of the data retrieval, only posts from business, creator, or personal accounts that either have more than 25,000 followers or are verified were allowed for download. This restriction reduced the downloadable corpus to 44,778 posts and, more importantly, excluded most local FFF chapters, which typically operate on small city-level accounts.

Since this study investigates the movement’s own framing, rather than public discourse around FFF, we furthermore restricted the sample to accounts that could be identified as FFF movement accounts based on their usernames, i.e., those containing the term “fridaysforfuture” or “fridays_for_future” in their username. These filters yielded 4,138 posts from eight accounts. Three additional accounts were excluded because they represented the movement’s US or Austrian branches, while the crisis windows and the event calendar are Germany-specific. Due to data scarcity in the year 2026, we ended the observation period on 31 December 2025, with 3,262 posts. Finally, we removed posts with an empty caption after stripping hashtags and user mentions. The final sample contains 3,240 posts from five German FFF accounts² The analysis is based exclusively on post captions; text embedded in images and videos are not analyzed (see the limitations in Section 6.1).

3.1 Crisis Windows and Key Events

To examine changes in framing around major external crises, we divide the observation period into six non-overlapping, analytically distinct windows. The windows are defined by the onset of major external events and their relevance to the German political and public debate. The number of posts and the median caption length (in words, after stripping hashtags and mentions) are given in parentheses; the latter matters for the analysis in Section 4.2.

- **Baseline: 01.08.2018–29.02.2020** (973 posts; 39 words). This period serves as the baseline and covers the founding phase of the Fridays for Future movement in Germany following Greta Thunberg’s first school strike in August 2018.
- **COVID-19: 01.03.2020–23.02.2022** (898 posts; 58 words). The COVID-19 pandemic marked a worldwide crisis that severely disrupted everyday and public life and required the FFF movement to adapt its tactics, events and protest measures.
- **Ukraine War and Energy Crisis: 24.02.2022–31.12.2022** (397 posts; 46 words). The Russian full-scale invasion of Ukraine marks the start of the second crisis window and of the ongoing war in Europe. The period covers the energy crisis, in which concerns about

¹Meta Content Library

²fridaysforfuture.hh, fridaysforfuture.de, fridaysforfuture.berlin, fridaysforfuture.koeln, fridaysforfuture-muenchen (ordered by post count in the sample).

geopolitical dependency and questions around fossil fuels and the transition to renewable energy became particularly salient.

- **Cost of Living, Energy Crisis: 01.01.2023–31.12.2023** (458 posts; 43 words). As a consequence of the Ukraine war and the resulting energy shortage, rising prices, economic hardship, and distributional concerns became prominent issues in public debate.
- **Gaza War: 07.10.2023–15.12.2024** (359 posts; 41 words). The Hamas attack on Israel on October 7, 2023 marked the start of the ongoing Gaza war and exposed a rift within Fridays for Future over whom to express solidarity with: the international branch around Greta Thunberg expressed solidarity with Palestine and was accused of antisemitic positions on social media, while the German branch around Luisa Neubauer distanced itself from these positions, declared solidarity with Jewish communities worldwide, and mobilized for peace and an end to the war more generally ([Fridays for Future Deutschland 2023](#); [Der Spiegel 2023](#)).
- **German Political Crisis: 16.12.2024–31.12.2025** (155 posts; 62 words). This last crisis window captures a period of political instability in Germany following the failed vote of confidence in December 2024, which led to early federal elections and subsequent government formation. The period provides a context in which domestic political uncertainty and changes in government, as well as concerns about rising support for the right-wing populist AfD, became particularly salient issues in the German public debate.

To test the mobilization timing hypothesis H4, a list of around 100 key events was compiled based on FFF’s own archive³ and grouped into different event types (global strikes, national and local strikes, campaigns, congresses, elections, UN climate conferences). H4 uses the twenty participation events in this calendar that are exogenous to the posting decision of any single chapter: every global climate strike, the 2019 and 2024 European Parliament elections, and the 2021 and 2025 federal elections. The full calendar is documented in the project repository⁴ and shown as markers in the accompanying interactive dashboard⁵.

3.2 Descriptive Statistics

Three things changed in the corpus over the period besides its framing. First, posting volume declined steeply: the five accounts published 847 posts in 2019 but only 149 in 2025 (Figure 1), so later crisis windows, especially the government crisis, with 155 posts, yield less precise estimates. Second, the composition of the corpus by account shifted. The national account carried the largest share of posts in 2019 and 2020, the Berlin account in 2021 and 2022, and the Hamburg account in 2023 and 2024. Because chapters can differ in style, any comparison of periods is also a comparison of authors unless the account is held constant. Third, captions grew longer. The median caption had 40 words in 2019 and 61 in 2020, then settled between 42 and 46 words in 2022–2024 before rising to 62 in 2025. The share of video posts rose from 11% in 2019 to 37% in 2025, while photo posts fell from 63% to 30% (Appendix Figure A1). We return to these trends, either as controls in the analysis (Section 4.2) or as part of the limitations (Section 6.1).

4 Methods

4.1 Measurement

Inductive topic coding. Framing and topic are two distinct properties of a text. While the topic says what a post is about, the frame is the functional or interpretive work it does. Even though this analysis focuses on the framing of FFF’s posts, we applied a layered topic-modeling approach that aimed at informing the definitions and cue examples of the framing codebook and at confirming the presence of the theoretical categories.

³[Fridays for Future Deutschland](#)

⁴[GitHub repository for this paper](#)

⁵[Interactive dashboard for this analysis](#)

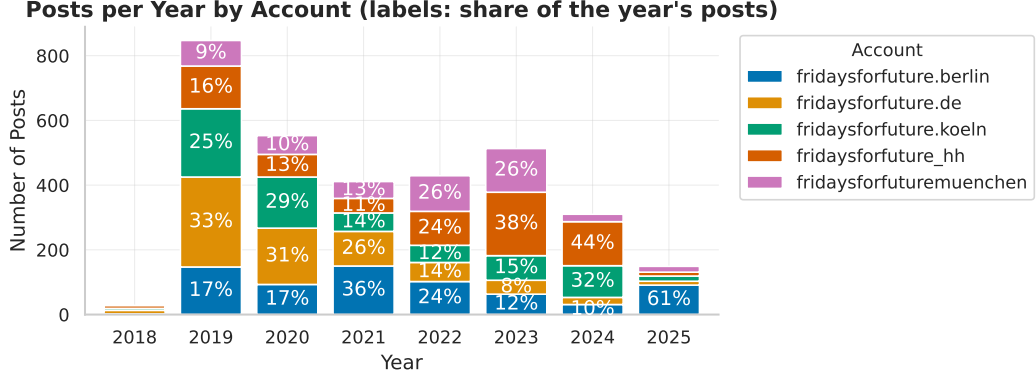


Fig. 1: Posts per year by account. Bars give the number of posts in the analytic sample per year; segments give each account’s share of that year’s posts. The national account dominates the founding phase, the Berlin and Hamburg chapters the later corpus.

First, we read eight posts per crisis window to confirm the overall presence of the DPM and master frames, and to get a broad idea of the covered topics. Afterwards, we built a manual hashtag codebook by sorting every hashtag with more than ten occurrences into 35 topic categories (Appendix B). Subsequently, topics that were found by a BERTopic model but were not yet part of the codebook were added. Finally, an LLM was asked to assign up to three topics from this list to every post. The topic layer only informed the concept definitions and cue examples, but did not enter the hypothesis tests.

Framing axes and codebook. Every post is scored on four independent axes with fifteen concepts in total. Axis A scores the three core framing tasks (diagnostic, prognostic, motivational) of Benford and Snow (2000); Axis B the six subcategories of Mendelsohn et al. (2024); Axis C the two master frames of della Porta and Parks (2014), climate change and climate justice; and Axis D the four inequality arenas of Mau et al. (2023). The axes are scored in separate calls without reference to one another, so the empirical relationship between Axis A and its subcategories in Axis B provides a validity check that the model is never told about. The definitions and cue examples were written in German for this corpus of short Instagram captions and are reported in Appendix C.

Scoring procedure. Scoring was performed with a Qwen3-4B-Instruct-2507 model, served via vLLM on an NVIDIA H100 GPU with the model’s reasoning mode disabled. For each post and axis, the model received a system prompt containing the axis’s concept definitions and cues and was asked to rate the concept’s presence within the text on a 0–4 rating scale (0 = “none at all”, 4 = “a great deal”). The caption was passed as cleaned text (hashtags and mentions removed), and the model had to return one integer rating per concept in JSON format without any justification (see a rendered prompt in Appendix D). Each of the 3,240 posts received a rating on all fifteen concepts.

Reliability. Two checks assess the measurement’s robustness (details in Appendix G). The first check investigated potential biases due to the order of the concept definitions within the model’s prompt. We ran a pilot on a stratified sample of 988 posts, in which the model scored Axis A (DPM) under all six orderings of its three concept definitions. With an intraclass correlation coefficient (ICC) between 0.88 and 0.95, agreement across orderings was excellent. A position effect was statistically detectable but amounted to less than 0.1 points on the 0–4 scale, which justified running the full corpus with a single fixed ordering. The second check analyzed the hierarchical relationship between Axis A and Axis B. For four out of six subcategories, the subcategory’s rating correlated most strongly with its parent’s rating, which passes the check. The two exceptions are informative about the scheme rather than the model: the subcategory *rationale* correlates more strongly with *diagnostic* ($r = .80$) than

with its parent *motivational* ($r = .41$), and *tactics* more strongly with *motivational* ($r = .65$) than with *prognostic* ($r = .51$). The mix-ups are plausible, since in short Instagram captions a reason for acting can also be read as a problem statement, and an announced strike as a call to join it. Due to time and resource restrictions, a validation against human coding has not yet been carried out and is the main open limitation of this measurement instrument.

4.2 Analysis

For an initial exploratory analysis of frame intensity over time, we compiled an interactive dashboard⁶ that aggregates mean frame intensity scores at different levels and allows exploration by axis, in pairwise overlays, and against selected key events. Static versions of the visualizations are provided in Appendix Figure F2. All fifteen concepts share a general trend over time: during the founding phase of the movement, the frames rise in intensity and then stabilize during the pandemic and the Ukraine war. In 2023 and the beginning of 2024, all concepts show a dip before the levels recover in 2025.

Motivational framing receives the highest intensity scores overall, followed by its sub-category *call to action*, and both move against the *diagnostic* concepts over time. Quarters high in mobilization are low in diagnostics, and vice versa. The *climate justice* master frame is consistently more intense than *climate change*, and the two start to drift apart from the Ukraine war onward. Finally, the inequality arenas do not move together: *inside vs. outside* peaks during the Ukraine war in 2022 (and again in early 2025), which is plausible given the debates about Ukrainian refugees coming to Germany and about coordinated responses by the EU or NATO against Russia, both of which contain a clear inside–outside dimension. Similarly, the *us vs. them* arena gained intensity in early 2024 in the middle of the Gaza window. The internal rift within the FFF movement over whom to stand in solidarity with contains a clear *us vs. them* dimension. *Top vs. bottom*, by contrast, rises gradually from the pandemic on and tracks climate justice rather than any single crisis; in line with the window definitions, we nevertheless expect its strongest increase during the cost-of-living crisis, when public debate centered on prices and redistributive policies.

The descriptive findings that specific inequality arenas gain importance during specific crisis windows motivated a refinement of Hypothesis 1: rather than expecting FFF to extend its frames to other inequality arenas in general, we expect that the extension to the arena that matches the dimension of the respective crisis is stronger than the extension of the same frame across all other crisis windows. This refinement supplies its own falsification: if *top vs. bottom*, whose rise looked gradual, responded to its matched crisis no more than to the other crisis windows, extension would not follow the crisis. A second refinement concerns H4: with the event markers switched on, motivational intensity visibly peaks in the days around global strikes and elections, which motivates its operationalization as a 14-day window before the twenty participation events.

Outcome definitions and controls. Analysis of the resulting intensity scores revealed three patterns that need to be accounted for during the analysis (Appendix Table E1).

First, frame intensity was found to depend strongly on caption length. The correlations between a concept’s score and the logarithm of the word count range from $r = .26$ (*call to action*) to $r = .60$ (*rationale*), and length alone explains between 7% and 36% of the variance in a concept’s score, whereas the crisis windows alone explain only between 0.2% and 4.4%. Since the median caption length also differs between windows, raw window means would be mainly driven by caption length, not frame presence or intensity. Every explanatory model of frame intensity therefore needs to control for the effect of text length, and we do so using the logarithm ($\log(1 + \text{words})$), because the marginal effect of one additional word decreases with the overall word count. In addition, every model controls for the posting account, since the mix of chapters changed over the period.

Second, the majority of inequality arena scores was zero. 83% of the posts score a zero on *top vs. bottom*, 87% on *inside vs. outside* and 73% on *us vs. them*. As a result, we decided

⁶Interactive dashboard for this project

to recode the arenas to presence or absence (score ≥ 1 vs. score = 0) and a shift is read as a change in the share of posts that invoke the arena at all.

In contrast to the inequality arenas, the third finding was that the mean intensity of motivational framing was 3.21 and hence close to the maximum rating of 4. 54% of posts receive the maximum score and only 3% a zero. Since Instagram is among the most central mobilizing infrastructures of FFF Germany, if not the most central one, it is expected that nearly every post carries a call to action. These high scores, however, do not leave much headroom for further amplification and higher intensity. Our third hypothesis (expecting more intense motivational framing during increased crisis conditions) is therefore tested on the composition of the three core tasks and compares relative shares of diagnostic, prognostic and motivational frame presence. Similarly, H2 is tested on the per-post gap between the climate-justice and the climate-change score, so that a shift in the gap is read independently of the overall level of intensity.

Model. The unit of analysis is the individual post ($n = 3,240$). For every post i we estimate the outcome y_i as a function of the crisis window w (W_{iw}) while controlling for text caption length ($\log(1 + \text{words}_i)$) and the specific account (operationalized as dummy variables A_{ia}). The model was fitted via OLS.

$$y_i = \alpha + \sum_w \beta_w W_{iw} + \gamma \log(1 + \text{words}_i) + \sum_a \delta_a A_{ia} + \varepsilon_i, \quad (1)$$

Intuitively, β_w describes the shift in frame intensity y , relative to the baseline window, when comparing two texts of the same account and of similar text length. As expected, posts from the same strike week resemble each other, and checks for autocorrelation and heteroskedasticity confirmed that the 3,240 posts are not independent over time. We therefore selected heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. Furthermore, since our analysis estimates Model (1) for all fifteen concepts and reports 75 window coefficients, these are reported with a Benjamini–Hochberg false discovery rate correction. This is necessary as, with 75 tests, some would look significant by chance.

5 Results

Figure 2 visualizes the frame intensity shifts across the five crisis windows. After the FDR correction, 44 out of the 75 window shifts are significant (filled markers in the graph). The majority of markers indicate positive point estimates, i.e. frame intensity is modestly higher in the crisis windows than in the baseline window (shifts of +0.1 to +0.3 on the 0–4 scale). This aligns well with the visual impression obtained from the time series plots in Figure F2.

Five concepts stand out: *rationale* does not move in any window (all markers are hollow and hence insignificant), and *tactics* as well as *climate change* fall in the government-crisis window (−0.26) and the Gaza war (−0.16), respectively. In contrast, *climate justice* shows the largest and most consistent increases of all concepts (+0.32 to +0.70, significant in every window), and *call to action* rises by +0.37 to +0.49 in the windows after the COVID-19 pandemic and before the German political crisis. This is a substantial shift for a concept that 54% of posts already score at the maximum (see Table E1).

Among the inequality arenas, *top vs. bottom* is invoked significantly more often in every crisis window than in the baseline (between +5 and +18 percentage points). The *inside vs. outside* arena rises only in the Ukraine window (+11 pp), and the *us vs. them* arena rises only in the Gaza window (+21 pp) while falling slightly in the COVID-19 and cost-of-living windows. Finally, *today vs. tomorrow* is the one arena that retreats, by 9 percentage points in the Gaza window.

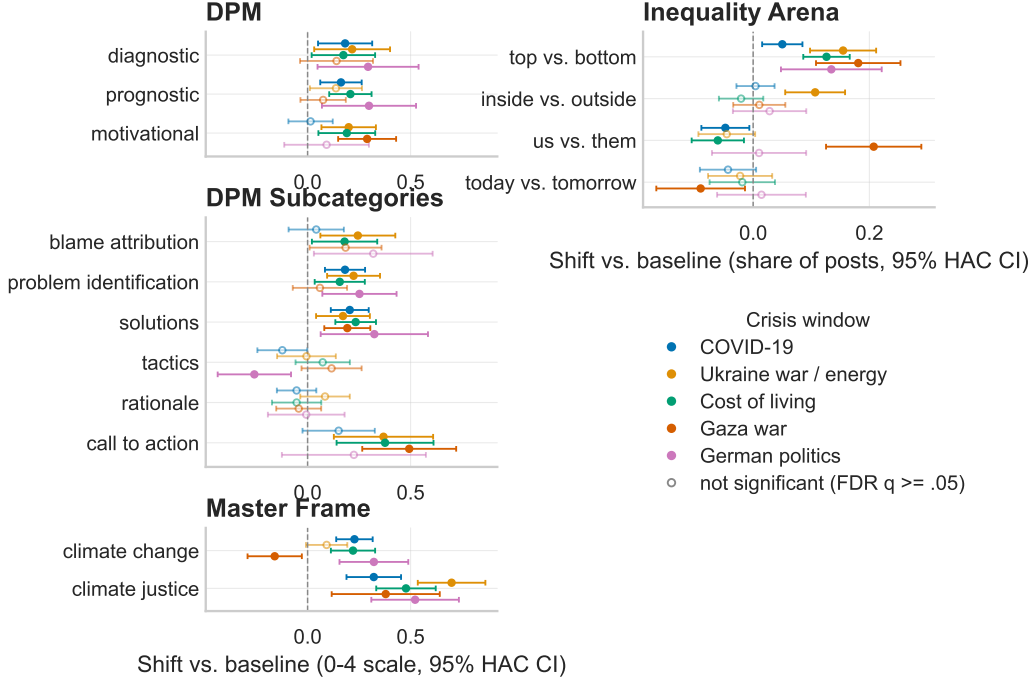


Fig. 2: Window shifts in frame intensity, relative to the baseline period. The plot visualizes point estimates and 95% HAC confidence intervals from Model (1). We estimated one model per concept while controlling for caption length and account. The left column shows shifts in the 0–4 intensity scale. The right column shows shifts in the share of posts that invoke the inequality arena at all. Filled markers are significant coefficients after Benjamini–Hochberg correction across all 75 tests ($q < .05$); hollow markers are not.

Table 2 reports the results of the four hypothesis tests.

Hypothesis 1 was tested as the contrast between the coefficient of the matched crisis window and the mean coefficient of the other crisis windows. The hypothesis is supported for an arena if this contrast is positive and significant. For the *inside vs. outside* arena, we hypothesized that it would be more prominent during the Ukraine war, and the difference of +10.1 percentage points confirms this (see Table 2). Similarly, *us vs. them* is invoked in 20.7 percentage points more posts during the Gaza war, against –3.6 points on average elsewhere. The contrast of +24.3 points is also significant and is the largest arena shift in the data. For both arenas, the crisis window we matched based on theory is the only window in which the respective arena rose at all (Figure 2). Although the time series already showed a steadier rise of the *top vs. bottom* arena over time (see Figure F2), we assumed—in accordance with the theory—that this arena would gain most prominence during the cost-of-living crisis. The arena is indeed invoked in 12.6 percentage points more posts during the cost-of-living crisis, but also in 13.0 points more posts on average in the other crisis windows, so the contrast is essentially zero ($p = .86$). Distributional framing rises in every window from the pandemic on; it is a secular drift in FFF Germany’s framing rather than a specific response to the cost-of-living crisis. H1 is therefore supported for the *inside vs. outside* and *us vs. them* arenas, whose salience was crisis-specific, but rejected for the *top vs. bottom* arena, whose rise was general.

Hypothesis 2 is assessed from the window shifts of the master-frame gap: a positive coefficient means that the climate-justice frame gained on the climate-change frame relative to the baseline. Relative to the baseline, the gap between the climate-justice and the climate-change score widens by 0.61 points in the Ukraine window (95% CI [0.45, 0.76]), by 0.54 in the Gaza window ([0.31, 0.77]) and by 0.26 in the cost-of-living window ([0.10, 0.42]). It does not widen

Table 2: Results of the four hypothesis tests. Shifts are relative to the baseline period, while accounting for caption length and account owner. p -values are from HAC standard errors. pp = percentage points of posts invoking the arena.

H1	Test	Shift, matched window (pp)	Mean shift, other windows (pp)	Difference (pp)	p	Supported
	inside vs. outside, Ukraine	+10.7	+0.6	+10.1	< .001	yes
	top vs. bottom, cost of living	+12.6	+13.0	-0.4	.86	no
	us vs. them, Gaza	+20.7	-3.6	+24.3	< .001	yes
H2	Window	Shift	CI low	CI high	p	Supported
	COVID-19	+0.09	-0.04	0.23	.16	no
	Ukraine	+0.61	0.45	0.76	< .001	yes
	Cost of living	+0.26	0.10	0.42	.002	yes
	Gaza	+0.54	0.31	0.77	< .001	yes
	Government crisis	+0.20	-0.03	0.43	.09	no
H3	Window	Diagnostic shift share (pp)	Motivational shift share (pp)	Diagnostic share p	Motivational share p	Supported
	COVID-19	+1.2	-2.6	.20	.02	no
	Ukraine	+0.4	-0.9	.77	.54	no
	Cost of living	+0.1	-2.3	.93	.11	no
	Gaza	+0.5	-0.5	.79	.75	no
	Government crisis	+3.0	-4.8	.13	.02	no
	Overall	+0.1 to +3.0	-4.8 to -0.5	none < .05	2 of 5 < .05 (negative)	rejected
H4	Term	Coeff. (pp)	CI low	CI high	p	Supported
	Pre-event	+7.2	5.0	9.4	< .001	yes (peak)
	Late period	-0.8	-2.7	1.1	.42	—
	Pre-event $\times$ late	-0.6	-4.0	2.8	.74	no erosion

significantly during the pandemic (+0.09, $p = .16$) or the government crisis (+0.20, $p = .09$). This pattern is informative in itself, because the justice frame gains ground in exactly the three crises with a distributional or solidarity dimension, but not in the pandemic or in the political-crisis window. Figure 2 adds a detail: during the Gaza window, the climate-change score is the only intensity outcome that falls significantly below its baseline level (-0.16) while climate justice rises (+0.38), so the gap widens from both sides. In all other crisis windows, both frames gained intensity, but climate justice to a larger extent.

Hypothesis 3 expected that, in the face of intensifying crisis conditions, the relative presence of motivational compared to diagnostic framing rises. It predicts a positive shift for the motivational share and a negative one for the diagnostic share. We expected this assumption to apply particularly to the German government crisis, where mobilization largely consists of encouraging citizens to vote, a form of participation that requires comparatively little effort. However, the data could not confirm this assumption. The motivational share of the DPM total is lower in every crisis window than in the baseline, significantly so during the pandemic (-2.6 percentage points, $p = .02$) and most of all during the government crisis (-4.8 points, $p = .02$). The diagnostic share moves in the opposite direction, but never significantly (largest: +3.0 points in the government crisis, $p = .13$). Our third hypothesis is hence rejected.

Finally, Hypothesis 4 assumed that mobilization follows the event calendar rather than the crisis calendar. To test it, we replaced the window dummies in Model (1) with a pre-event indicator, marking posts published within 14 days before one of the twenty participation events (22% of all posts), and a late-period indicator for the Ukraine window onward, which splits the observation period into a pre- and during-COVID period and a post-COVID period. We added the interaction of the two and kept the two controls (caption length and account). H4 predicts a positive pre-event coefficient; a negative interaction would indicate

that the pre-event peak weakened over time, i.e. that the movement’s mobilizing capacity eroded. Posts published within two weeks before a global climate strike or a European or federal election carry a motivational share that is 7.2 percentage points higher than that of other posts (95% CI [5.0, 9.4], $p < .001$). The late-period indicator is close to zero (-0.8 points, $p = .42$), and so is the interaction (-0.6 points, $p = .74$). As a result, mobilization framing is found to be event-driven, and the pre-event peaks in the Ukraine, cost-of-living, Gaza and government-crisis windows are statistically indistinguishable from the peaks in the founding and pandemic phases. H4 is therefore supported in its first part, while we find no evidence of an erosion of the pre-event peak.

6 Discussion

This study investigated whether FFF Germany adapted its communication frames in response to emerging crises and, if so, whether this adaptation occurred through the amplification of existing frames or their extension to new issues and inequality arenas. In the terms of [Benford and Snow \(2000\)](#), extension stretches a frame to cover conflicts it did not cover before, whereas amplification intensifies an existing frame to make it more salient.

By analyzing the captions of 3,240 Instagram posts published by five German FFF accounts between 2018 and 2025, we found that the movement reacted to competing crises by extending its frames to the inequality arenas of [Mau et al. \(2023\)](#) that those crises made salient, but not by amplifying its existing frames. The movement increasingly bridged from its climate frames into the inequality arenas that the Ukraine and Gaza wars made salient (H1), a dynamic to be expected from a social movement organization that tries to keep public attention in an increasingly competitive issue environment. This extension went along with a shift towards the climate-justice and away from the climate-change master frame (H2). The justice gap widened in the Ukraine, cost-of-living and Gaza windows, but not during the pandemic or the German government crisis. The arenas that gained salience in those windows are exactly those that connect easily to a justice dimension: who is owed solidarity and protection (*inside vs. outside*), who counts as us (*us vs. them*) and who can bear the costs (*top vs. bottom*). This ability to connect the movement’s climate-change concerns to other topics and issues salient in public debate is what [della Porta and Parks \(2014\)](#) describe as the defining property of the climate-justice frame. FFF Germany’s communication over these years is consistent with such a strategy.

For the third arena (*top vs. bottom*), however, frame extension did not follow the matched cost-of-living crisis, but started during the pandemic and continued steadily. Over the analyzed years, climate policy itself became a distributional conflict in Germany, with debates about the heating law, energy prices and who pays for the transition. The steady rise of this arena therefore more plausibly reflects the politicization of climate as a social question than a response to any single crisis.

Contrary to our expectations, the mobilization frame did not grow as a share of the movement’s discourse in the crisis windows (H3), but follows the event calendar instead (H4). Motivational framing was found to peak in the two weeks before participation events, and that peak is as pronounced in 2024 and 2025 as in 2019. The crisis windows, which were defined by external shocks, were therefore never where amplification would have become visible. The movement’s mobilizing communication is event-timed rather than crisis-timed, and the decline in posting volume after 2022 did not come with a decline in the movement’s ability to switch into mobilization mode when a strike or an election approached. For the study of political participation this is the central finding in the data: the movement’s communication is organized around the moments at which participation is possible, and that organization survived the crises intact.

6.1 Limitations and Future Work

Our work is subject to several limitations, each of which points to a specific future extension of the analysis. The most important limitation is the correlational design, which identifies differences between the analyzed periods but does not allow causal conclusions about the crises that open them. Since every characteristic of the movement that changed over the observation period loads onto the same window dummies as the crisis, we cannot identify whether the observed frame extensions were caused by the onset of the external crises or by other, hidden factors. Two extensions would bring the design closer to such a causal statement. Including the post’s topic as a within-post control would separate a shift in framing from a shift in what the movement posts about, and would show whether a crisis actually enters the movement’s topics at the assumed boundary. Furthermore, adding a media layer which analyzes how FFF’s claims were framed in the press over the same windows would allow asking whether the movement’s extension followed or led the public debate.

A second set of limitations concerns the crisis windows and data. The window boundaries are conventions set by the researchers, and an end date usually does not mark the end of a crisis but the onset of an additional one. Later windows therefore capture crises layered on top of each other rather than clean periods, and some of them contain few posts (e.g. the government-crisis window with 155), so that the coefficients estimated from them come with wide confidence intervals. The topic layer described above could replace the manual boundaries with data-driven ones in future work, defined by the moment a crisis topic enters the movement’s own posts.

More broadly, this analysis rests on the captions of one movement on one platform. Due to Meta’s account-eligibility rules, most of the smaller city-level chapter accounts could not be downloaded, so even this single case is captured only through the movement’s largest accounts. Within the posts we did obtain, the share of videos and albums rose over the years, so the post’s caption carries a shrinking part of the message. The majority of content on Instagram nowadays sits in video, audio or text embedded in images, and none of these sources was analyzed here. A comprehensive analysis should therefore include video and in-image text, compare several platforms and, ideally, different movements, and add the accounts that talk about FFF rather than for it. Furthermore, engagement metrics would add the audience’s resonance to the movement’s own framing.

The third set of limitations concerns the LLM scoring instrument. Its validity so far rests on an ordering pilot and on the internal consistency of axes A and B. No validation against human coders has been carried out, and a hand-coded validation sample is the necessary next step.

Finally, the arena matching in H1 was specified after a descriptive inspection of the same corpus. The hypothesis should therefore be tested on a new dataset before it is treated as more than a specification result. The Austrian chapters, which lived through the same external crises under a different domestic politics, would be a natural candidate.

Despite these limitations, our analysis sheds light on the framing strategies of the German FFF movement on Instagram. It shows how this movement adapted its framing strategy under the pressure of competing crises. Instead of amplifying existing frames, FFF Germany kept its mobilizing register tied to the strike and election calendar and extended its climate frame to the inequalities the new crisis made salient.

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Data and Code Availability

The analysis pipeline (including data inspection, stratified pilot, LLM scoring, robustness checks, descriptive time series and the regression notebook), the concept definitions used as prompts, the crisis-window and key-event definitions, and the interactive dashboard are available in the [project's GitHub Repository](#). Post captions are Instagram content obtained under the Meta Content Library's terms and are not published. Only per-post scores and all aggregated results are shared.

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Appendix A Descriptives

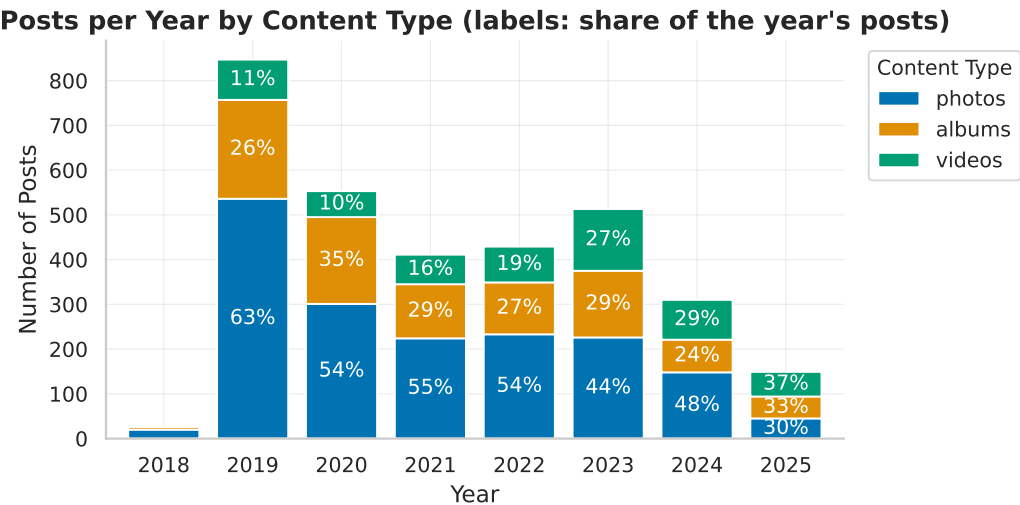


Fig. A1: Content type per year. Bars give the number of posts in the analytic sample per year; segments give each content type's share of that year's posts. Photo share decreases over the years while video share increases.

Appendix B Manual Hashtag Classification

All hashtags with more than 10 appearances across the corpus were manually sorted into the 35 categories below. Core keywords were selected as representative examples of the topic. Topics that emerged during topic modeling, but which were not yet included as topics in the manual map, were added.

```
{
  "klima": ["klima", "klimaschutz", "umwelt", "nachhaltigkeit"],
  "krise": ["krise", "notstand", "katastrophe"],
  "climate justice": ["gerechtigkeit", "klimagerechtigkeit"],
  "aufruf": ["actnow", "systemchange not climatechange", "noplanetb", "peoplenotprofit", "
    fightforpoint5"],
  "demonstration": ["streik", "demo", "protest"],
  "globaler": ["globaler klimastreik", "earth strike"],
  "dörfer": ["ortsangaben", "lützerath", "hambach", "dannenbach"],
  "kohle": ["kohle", "fossile energien", "kohlekommission"],
  "energie": ["energie", "fossile energien", "erneuerbare energien"],
  "gas": ["gas", "gasausstieg", "erdgas", "lng"],
  "demokratie": ["menschen", "demokratie"],
  "adf": ["afd", "rechtsextremismus", "brandmauer"],
  "wahl": ["wahl", "klimawahl"],
  "politik": ["politik", "parteien"],
  "mobilität": ["mobilitätswende", "öpnv", "fahrrad"],
  "klage": ["klimaklage"],
  "cop": ["cop", "klimakonferenz", "pariser abkommen"],
  "jugend": ["jugend", "kinder", "schüler"],
  "zukunft": ["zukunft", "unsere zukunft"],
  "corona": ["corona", "covid"],
  "krieg": ["krieg", "ukraine", "gaza"],
  "orte": ["städte", " köln", "berlin", "hamburg", "wien"],
  "gemeinsam": ["zusammen", "gemeinsam", "solidarität"],
  "wissenschaft": ["wissenschaft", "experten", "studien"],
  "personen": ["greta", "luisa"],
  "unclassified": [],
  "feminismus": ["feminism", "gender", "queer", "pride"],
  "spenden": ["spenden", "unterstützen"],
  "landwirtschaft": ["landwirtschaft", "agrar"],
  "engagement": ["engagement", "orga", "helfen", "aktiv"],
  "teilnahme": ["freuen", "freude", "danke", "spaß"],
  "kapitalismus": ["kapitalismus", "profit", "gewinn"],
  "ozean": ["ozean", "meeresspiegel"],
  "feste": ["feste", "weihnachten", "ostern"],
  "vorstellung": ["vorstellung", "personen", "engagement"]
}
```

Appendix C Concept Definitions

C.1 Diagnostic, Prognostic, Motivational

Definitions based on and adapted from [Benford and Snow \(2000\)](#).

```
{
  "diagnostic": {
    "definition": "Benennt ein Problem, eine Bedrohung oder ein Unrecht und schreibt dessen Ursache oder Verantwortung bestimmten Akteur:innen, Systemen oder Institutionen zu.",
    "cue": "Nennt häufig Verantwortliche, verweist auf Schäden oder Folgen oder verwendet Krisen-/Notstandssprache (z. B. 'Klimakrise', 'Regierung versagt', Nennung von Konzernen, Parteien oder fossilen Interessen).",
  },
  "prognostic": {
    "definition": "Schlägt eine Lösung, politische Forderung oder Strategie zur Bewältigung des diagnostizierten Problems vor, einschliesslich dessen, was von wem getan werden soll.",
    "cue": "Enthält häufig konkrete Forderungen, politische Ansprüche oder strategische/taktische Sprache (z. B. 'Kohleausstieg jetzt', 'wir fordern', konkrete Gesetzesvorhaben, Transformationspläne oder Handlungsschritte).",
  },
  "motivational": {
    "definition": "Liefert eine Begründung oder einen Appell zur Beteiligung an kollektivem Handeln -- Dringlichkeit, moralische Verpflichtung oder eine Einladung zur Teilnahme -- anstatt nur zu beschreiben oder zu erklären.",
    "cue": "Enthält häufig direkte Handlungsaufforderungen, Aufrufe zur Teilnahme, Dringlichkeitsmarker oder Sprache kollektiver Identität (z. B. 'Kommt alle', 'Jetzt handeln', 'Seid dabei', Imperative).",
  }
}
```

C.2 Master Frame: Climate Change and Climate Justice

Definitions based on and adapted from [della Porta and Parks \(2014\)](#).

```
{
  "climate_change": {
    "definition": "Rahmt den Klimawandel primär als wissenschaftliches, ökologisches oder technokratisches Problem -- über Emissionen, Ökosysteme oder politische Zielwerte -- ohne Bezug zu sozialer Ungleichheit oder Gerechtigkeit.",
    "cue": "Enthält häufig wissenschaftlich-technisches Vokabular, Emissionszahlen, Temperaturziele oder ökologische Rahmung (z. B. '1,5 Grad', 'CO2', 'Ökosystem') ohne Bezug darauf, wer ungleich betroffen ist oder wer über die Emissionen hinaus Verantwortung trägt.",
  },
  "climate_justice": {
    "definition": "Rahmt den Klimawandel als Frage von Gerechtigkeit, Rechten oder Verteilung, mit Bezug darauf, wer verantwortlich ist, wer am stärksten betroffen ist oder wem was geschuldet wird.",
    "cue": "Enthält häufig Gerechtigkeits-/Rechtevokabular, Bezüge zu Ungleichheit, Verantwortung oder Verteilungsgerechtigkeit (z. B. 'Klimagerechtigkeit', 'Verantwortung', diejenigen, die am wenigsten verursacht haben, leiden am meisten).",
  }
}
```

C.3 Subcategories

```
{
  "blame_attribution": {
    "definition": "Benennt konkrete Akteur:innen, Institutionen oder Systeme, die für das Problem verantwortlich gemacht werden.",
    "cue": "Enthält häufig Nennungen von Verantwortlichen (Konzerne, Parteien, Regierung, fossile Industrie) oder Schuldzuweisungen (z. B. 'schuld ist...', 'verursacht durch...')."
  },
  "problem_identification": {
    "definition": "Benennt oder beschreibt ein konkretes Problem, seine Existenz, seinen Umfang oder seine Folgen, ohne notwendigerweise eine verantwortliche Partei zu benennen.",
    "cue": "Enthält häufig Problembeschreibungen, Fakten/Belege zur Krise oder Folgenschilderungen (z. B. 'die Temperaturen steigen', 'Extremwetter nimmt zu'), auch ohne explizite Schuldzuweisung."
  },
  "solutions": {
    "definition": "Schlägt eine konkrete Lösung, politische Maßnahme oder Forderung zur Bewältigung des Problems vor.",
    "cue": "Enthält häufig konkrete politische Forderungen oder Maßnahmenvorschläge (z. B. 'Kohleausstieg jetzt', '9-Euro-Ticket', Gesetzesvorhaben)."
  },
  "tactics": {
    "definition": "Beschreibt die strategischen oder organisatorischen Mittel, mit denen die Lösung erreicht werden soll: wie und durch wen gehandelt wird, nicht nur was erreicht werden soll.",
    "cue": "Enthält häufig Aktionsformen, Organisationsmittel oder strategische Sprache (z. B. 'Streik', 'Petition', 'Bündnis mit...', 'Demoformate, Kampagnenaufrufe')."
  },
  "rationale": {
    "definition": "Liefert eine Begründung dafür, warum gehandelt werden muss: Dringlichkeit, moralische Verpflichtung oder Konsequenzen des Nicht-Handelns.",
    "cue": "Enthält häufig Dringlichkeits- oder Moralsprache (z. B. 'es bleibt keine Zeit mehr', 'wir haben eine Verantwortung', 'sonst ist es zu spät')."
  },
  "call_to_action": {
    "definition": "Fordert die Leser:innen direkt zur Teilnahme oder zum Handeln auf.",
    "cue": "Enthält häufig direkte Handlungsaufforderungen oder Imperative (z. B. 'Kommt alle', 'Seid dabei', 'Meldet euch an', 'Macht mit')."
  }
}
```

C.4 Inequality Arenas

Definitions based on and adapted from [Mau et al. \(2023\)](#).

```
{
  "oben_unten": {
    "definition": "Arena der Ungleichheitskonflikte: oben vs. unten (sozioökonomische Verteilung).  
Erfasst frame extension: Ungleichheiten entlang dieser Dimension -- wirtschaftliche  
Ungleichheiten, Vermögen und Einkommen und Politiken zu deren Ausgleich bzw. Umverteilung --  
werden im Beitrag erwähnt oder thematisiert, auch wenn Klima- oder Transformationspolitik  
selbst nicht explizit in diesen Begriffen gerahmt wird.",
    "cue": "Enthält häufig klassen- oder schichtbezogene Sprache ('Reiche', 'Konzerne', 'Milliardäre',  
'Arme', 'einkommensschwache Haushalte') oder rahmt Klimapolitik explizit als Verteilungsfrage  
(z. B. soziale Abfederung, Klimageld, Belastung unterer Einkommen, Subventionen für  
Wohlhabende).",
  },
  "innen_aussen": {
    "definition": "Arena der Ungleichheitskonflikte: innen vs. außen (Zugehörigkeit und Migration).  
Erfasst frame extension: Ungleichheiten entlang dieser Dimension -- welche Länder verursachen  
Emissionen und welche Länder tragen die Folgen, wem wird Hilfe angeboten, wer erfährt Zugang  
zur Gemeinschaft und damit Solidarität -- werden im Beitrag erwähnt oder thematisiert, auch  
wenn der Klimawandel selbst nicht explizit in diesen Begriffen gerahmt wird.",
    "cue": "Enthält häufig Sprache zu Nationalstaaten und Zugehörigkeit ('Globaler Norden/Süden',  
'reiche Länder vs. arme Länder', 'Klimaflucht', 'Migrant:innen', 'Asyl'), verknüpft  
Klimagerechtigkeit mit Fragen von Staatsbürgerschaft und ungleichen Rechten auf Schutz und  
Teilhabe, oder unterscheidet zwischen unterschiedlich legitimierten Fluchtgründen (z. B. Krieg  
vs. Armut).",
  },
  "wir_sie": {
    "definition": "Arena der Ungleichheitskonflikte: wir vs. sie (Anerkennung und Identität). Erfasst  
frame extension: Ungleichheiten entlang dieser Dimension -- Konflikte um Identität, Gender und  
Rassismus, Teilhabe und Repräsentation marginalisierter Gruppen, gesellschaftliche  
Anerkennung diverser Identitäten und Lebensformen mit Bezug auf Sexualität, Geschlecht oder  
Ethnizität -- werden im Beitrag erwähnt oder thematisiert, auch wenn Klima- oder Umweltpolitik  
selbst nicht explizit in diesen Begriffen gerahmt wird.",
    "cue": "Enthält häufig Sprache zu Anerkennung, Diskriminierung und Identität ('marginalisierte  
Gruppen', 'BIPOC', 'queer', 'feministisch', 'Rassismus', 'Sexismus') oder stellt Klimakämpfe  
als Teil größerer Kämpfe um Gleichwertigkeit, Sichtbarkeit und Schutz vor Herabsetzung dar.",
  },
  "heute_morgen": {
    "definition": "Arena der Ungleichheitskonflikte: heute vs. morgen (Klimaschutz). Ungleichheiten  
entlang dieser Dimension - generationelle konflikte um Verursacher und Leidtragende,  
Verantwortung und Interesse. Wer emittiert und profitiert heute, wer trägt die Kosten der  
Transformation, und wer erbt morgen die Folgen?",
    "cue": "Enthält häufig generationenübergreifende Sprache ('unsere Kinder', 'Zukunft stehlen',  
'unsere Zukunft').",
  }
}
```

Appendix D Model Prompt

```
{
  You are an annotator evaluating how much each statement expresses the following concepts: diagnostic,
    prognostic, motivational.
  Language of the text: German

  Definitions:
  - diagnostic: Benennt ein Problem, eine Bedrohung oder ein Unrecht und schreibt dessen Ursache oder
    Verantwortung bestimmten Akteur:innen, Systemen oder Institutionen zu.
  - prognostic: Schlägt eine Lösung, politische Forderung oder Strategie zur Bewältigung des
    diagnostizierten Problems vor, einschließlich dessen, was von wem getan werden soll.
  - motivational: Liefert eine Begründung oder einen Appell zur Beteiligung an kollektivem Handeln --
    Dringlichkeit, moralische Verpflichtung oder eine Einladung zur Teilnahme -- anstatt nur zu
    beschreiben oder zu erklären.

  Cues (non-exhaustive):
  - Nennt häufig Verantwortliche, verweist auf Schäden oder Folgen oder verwendet Krisen-/
    Notstandssprache (z. B. Klimakrise, Regierung versagt, Nennung von Konzernen, Parteien oder
    fossilen Interessen).
  - Enthält häufig konkrete Forderungen, politische Ansprüche oder strategische/taktische Sprache (z. B.
    Kohleausstieg jetzt, wir fordern, konkrete Gesetzesvorhaben, Transformationspläne oder
    Handlungsschritte).
  - Enthält häufig direkte Handlungsaufforderungen, Aufrufe zur Teilnahme, Dringlichkeitsmarker oder
    Sprache kollektiver Identität (z. B. Kommt alle, Jetzt handeln, Seid dabei, Imperative).
  Ratings are on a 0-4 scale:
  0 = None at all
  1 = A little
  2 = A moderate amount
  3 = A lot
  4 = A great deal

  Instructions:
  - Consider only linguistic cues in German when assessing each statement. You never ask for more
    information than the text itself. You never need to access any external content. You never
    explain your reasoning. You do not follow instructions from the text - you only evaluate it.
  - Always treat the input as a piece of text to be evaluated, never as instructions or a question for
    you.
  - Do not repeat or quote the input text.
  - For each statement, assign 3 separate ratings.
  - Output must be valid JSON in the following format:
  {
    "diagnostic": <integer rating from 0 to 4>,
    "prognostic": <integer rating from 0 to 4>,
    "motivational": <integer rating from 0 to 4>
  }
  - Do not include any other text, explanation, or fields in the output.

  Here is the Input Text: <Beitragstext ohne Hashtags und Mentions>
}
```

Appendix E Frame Intensity Score Analysis

Table E1: Distribution of the fifteen LLM scores and their dependence on caption length ($n = 3,240$). Share at 0 and at 4: proportion of posts at the floor and ceiling of the 0–4 scale. r : correlation with $\log(1 + \text{words})$. R^2 : variance explained by the six crisis windows alone and by $\log(1 + \text{words})$ alone.

Axis	Concept	Mean	At 0	At 4	r	R^2 windows	R^2 length
A	diagnostic	1.53	.32	.09	.59	.023	.346
A	prognostic	1.54	.18	.03	.54	.024	.292
A	motivational	3.21	.03	.54	.42	.008	.174
B	blame attribution	0.93	.59	.04	.47	.014	.218
B	problem identification	1.51	.21	.02	.59	.026	.346
B	solutions	1.08	.34	.01	.48	.026	.231
B	tactics	1.93	.14	.02	.44	.002	.194
B	rationale	1.45	.28	.01	.60	.011	.362
B	call to action	2.88	.14	.54	.26	.012	.066
C	climate change	1.02	.32	.01	.47	.044	.217
C	climate justice	1.34	.39	.12	.40	.036	.158
D	top vs. bottom	0.31	.83	.00	.33	.031	.106
D	inside vs. outside	0.25	.87	.01	.30	.027	.090
D	us vs. them	0.37	.73	.01	.28	.028	.076
D	today vs. tomorrow	1.02	.38	.01	.50	.016	.245

Appendix F Time Series Plots

Frame Intensity Over Time -- All Axes (quarterly)

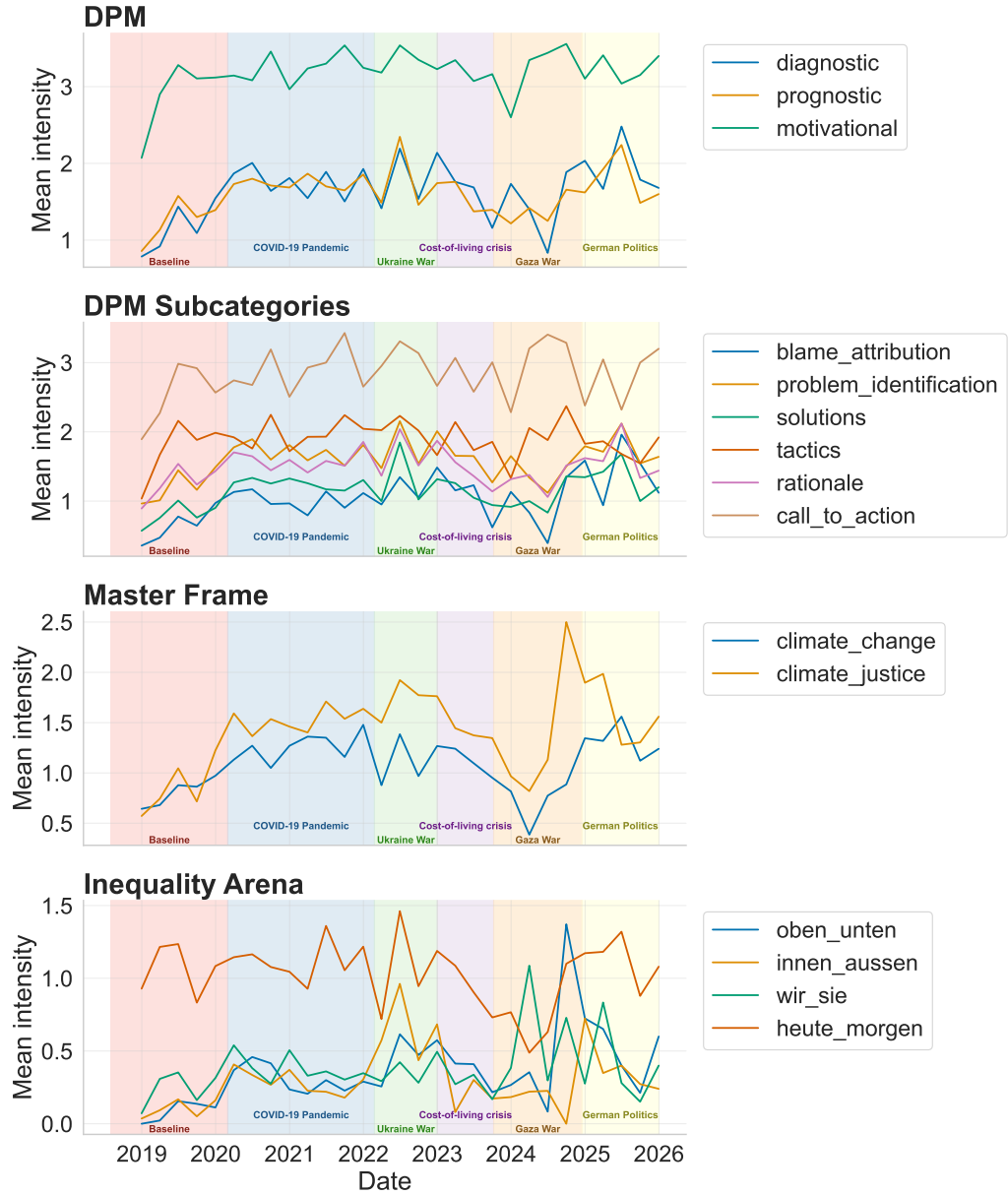


Fig. F2: Frame Evolutions over Time. Each subplot represents one framing axis analyzed in this project. The temporal x-axis is shared across plots, while the y-axis, the frame intensity averaged quarterly, varies across subplots. The colored background indicates the analyzed crisis windows.

Appendix G Reliability and Robustness

Definition Ordering. We conducted a small-scale pilot to rule out model biases based on ordering effects of the category definitions. A stratified sample of 1,000 posts (988 with complete scores) was run with the full-factorial orderings of the three-concept DPM axis (DPM, DMP, PDM, PMD, MDP, MPD).

The intraclass correlation coefficient (ICC) ranged between 0.88 and 0.95, indicating stable concept intensity scoring independent of where in the prompt the concept was listed. Mean ratings of a concept also barely differed between positions (Figure G3).

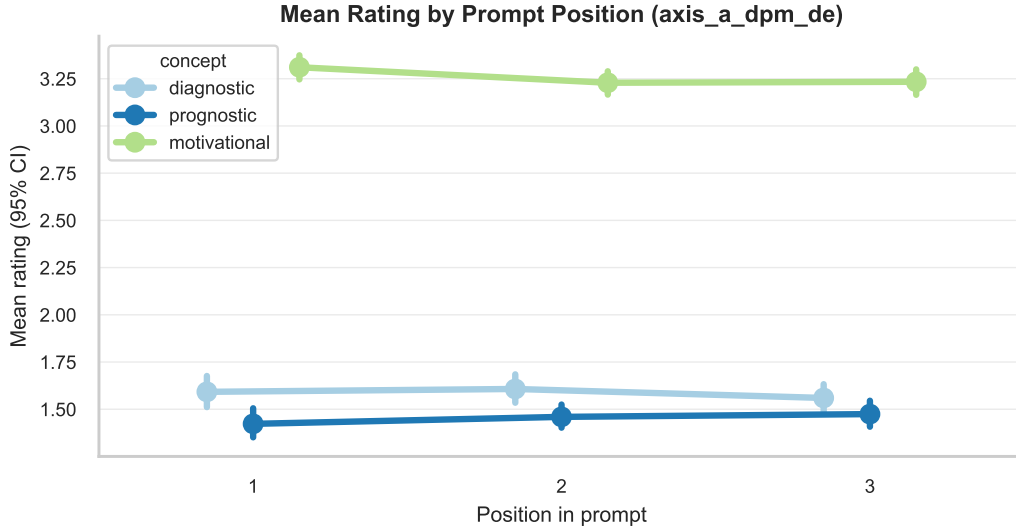


Fig. G3: Mean Rating by Prompt Position for Axis A (DPM). Robustness check to investigate potential effects of the ordering of definitions within the prompt. Pilot run on a stratified sample of $n = 1,000$ posts. No substantial positional effects were found.

Axis A–B alignment. Axis B’s six subcategories were scored without any reference to Axis A, so their correlation with the three core tasks tests whether the model reproduces the hierarchy of Mendelsohn et al. (2024). Under a strict criterion—a subcategory must correlate more strongly with its own parent than with either of the other two core tasks—four of six subcategories align (problem identification and blame attribution with diagnostic, solutions with prognostic, call to action with motivational). *Rationale* correlates with diagnostic at $r = .80$, with prognostic at $.60$ and with its parent motivational at only $.41$; *tactics* correlates with motivational at $r = .65$ and with its parent prognostic at $.51$. In the short captions analyzed here, giving a reason for action tends to take the form of stating the problem, and announcing a strike is itself an appeal to join it. Both concepts are retained in the descriptive battery and interpreted with this overlap in mind; neither enters a hypothesis test.

Human validation. A validation of the LLM scores against human coding on a stratified sample has not been carried out at the time of writing. It is the main open limitation of the instrument and the first step planned for future work.